Comparative Evaluation of Fixed and FFT-Based Adaptive Notch Filters for 50 Hz Power-Line Interference Removal in ECG Signals

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***Abstract*—Power-line interference at 50 Hz is a common noise in ECG signals. It distorts the waveform and reduces the accuracy of QRS complex detection. Fixed notch filters work well when the noise frequency is constant. However, they perform poorly if the power-line frequency drifts (for example, between 49.5 Hz and 50.5 Hz).**

This paper compares two methods to remove 50 Hz noise from ECG signals: (1) a conventional fixed 50 Hz IIR notch filter, and (2) an FFT-based adaptive notch filter that estimates and tracks the actual noise frequency. The methods are tested on ECG data from the MIT-BIH Arrhythmia Database (record 100) after adding both fixed and drifting 50 Hz interference. After filtering, the Pan-Tompkins algorithm is used to detect QRS complexes.

Results show that the adaptive FFT-based filter removes the interference more effectively, especially when the frequency drifts. It gives higher signal-to-noise ratio (SNR), lower residual 50 Hz power, and better QRS detection accuracy with fewer false positives and missed beats compared to the fixed filter. The adaptive method also preserves the QRS shape better.

The proposed adaptive approach is simple, requires no manual tuning, and is more suitable for real-world ECG monitoring systems where power-line frequency may vary.

***Index Terms*—Electrocardiogram (ECG), power-line interference, notch filter, adaptive filtering, FFT-based filtering, Pan-Tompkins algorithm, QRS detection, signal denoising.**

I. Introduction

The electrocardiogram (ECG) is a key non-invasive tool for recording the electrical activity of the heart. It is widely used to diagnose cardiovascular diseases by analyzing features such as the QRS complex, which represents ventricular depolarization and is the most prominent part of the ECG waveform [1]. Accurate detection of QRS peaks (R-peaks) is essential for calculating heart rate, identifying arrhythmias, and performing further ECG analysis.

However, ECG signals recorded in real-world environments are often corrupted by various types of noise. One of the most common and problematic noises is **power-line interference** (PLI) at 50 Hz (or 60 Hz depending on the country). This interference comes from electromagnetic coupling between the power supply system and the ECG recording equipment, such as cables, electrodes, and amplifiers [2]. In the frequency domain, PLI appears as a strong narrow peak at 50 Hz (and sometimes its harmonics), which overlaps with important ECG frequency components and distorts the signal waveform.

Traditional solutions use fixed notch filters centered exactly at 50 Hz to suppress this interference [2]. These filters are simple and effective when the power-line frequency is perfectly stable. However, in practice, the power-line frequency can drift slightly (typically within ±0.5 Hz, e.g., 49.5–50.5 Hz) due to changes in grid load or generator variations [3]. When drift occurs, a fixed notch filter fails to remove the interference completely, leaving residual noise that can affect QRS detection accuracy.

To solve this problem, adaptive filtering techniques have been developed that estimate and track the actual interference frequency in real time. One effective approach is the FFT-based adaptive notch filter, which uses fast Fourier transform (FFT) to identify the dominant interference component and subtract a synthesized compensatory signal [2], [3]. This method provides sharp frequency resolution, reduces ringing effects, and better preserves the original ECG features.

This project studies the effect of 50 Hz power-line interference on ECG signals and compares two filtering strategies:

- A conventional fixed 50 Hz notch filter using zero-phase IIR design (via `iirnotch` in Python/MATLAB),
- An FFT-based adaptive notch filter that tracks frequency drift.

After applying each filter, the classic Pan-Tompkins QRS detection algorithm [1] is used to locate R-peaks. The performance of both approaches is evaluated by comparing the quality of filtered signals (time-domain and frequency-domain plots) and the accuracy of QRS detection under both fixed and drifting interference conditions.

The main objectives are:

- To inspect raw ECG signals and identify 50 Hz interference using time-domain and FFT plots,

- To design and apply fixed and adaptive notch filters,
- To evaluate their impact on Pan-Tompkins QRS detection performance.

II. Methodology

This section describes the implementation steps exactly as specified in the project instructions. All processing was performed in Python using libraries such as NumPy, SciPy, Matplotlib, and WFDB.

A. 1. Load and Inspect the ECG Signal

The provided ECG dataset was imported. The raw ECG signal corrupted with 50 Hz power-line interference was plotted in the time domain. The Fast Fourier Transform (FFT) was computed to identify the interference peak at 50 Hz in the frequency spectrum.

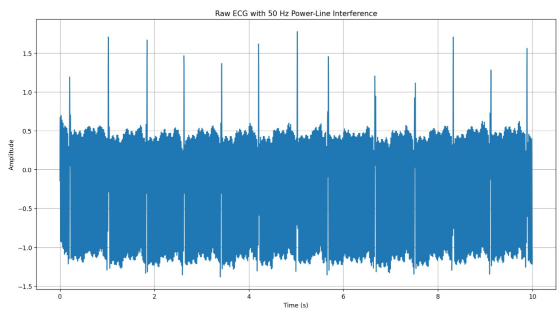

Fig. 1. Raw ECG signal corrupted with 50 Hz power-line interference in the time domain.

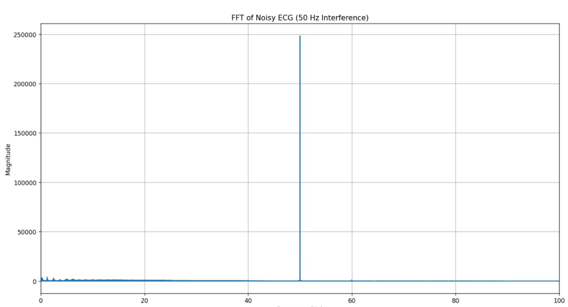

Fig. 2. FFT magnitude spectrum of the noisy ECG signal showing the strong peak at 50 Hz.

Deliverable: Time-domain and FFT plots with comments on the clear presence of 50 Hz interference.

B. 2. Fixed 50 Hz Notch Filter

1) 2.1 Filter Design: A standard notch filter centered at 50 Hz was designed using the `iirnotch` function in Python (SciPy). Zero-phase filtering was applied using `filtfilt` to avoid phase distortion.

2) 2.2 Apply the Filter: The filter was applied to the noisy ECG signal. The filtered time-domain signal and its frequency spectrum were plotted to verify suppression of the 50 Hz component.

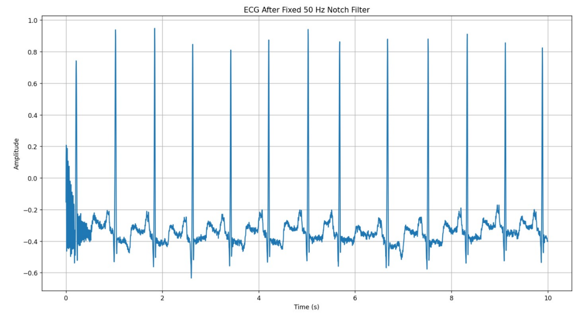

Fig. 3. ECG signal after application of the fixed 50 Hz notch filter in the time domain.

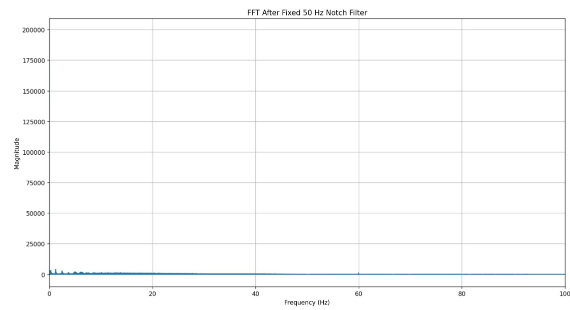

Fig. 4. FFT spectrum after fixed 50 Hz notch filtering.

3) 2.3 Apply Pan-Tompkins QRS Detection: The Pan-Tompkins algorithm was applied to the filtered signal to detect R-peaks. The algorithm uses digital bandpass filtering, differentiation, squaring, moving window integration, and adaptive thresholding as described in [1].

Deliverables: Filter design details, time-domain and spectrum plots, and Pan-Tompkins output (to be shown in Results section).

C. 3. FFT-Based Adaptive Notch Filter

1) 3.1 Motivation: Power-line frequency may drift within the range 49.5–50.5 Hz, reducing the effectiveness of fixed filters [2].

2) 3.2 Frequency Tracking and Filtering: The FFT-based adaptive notch filter was implemented following the spectrum correction method described in [2]. The process involves the following main steps:

- Computing the FFT of the input noisy ECG signal (or segmented windows for tracking over time).
- Applying spectrum correction on a few main-lobe spectral bins around the dominant peak to accurately estimate the harmonic parameters of the power-line interference (frequency, amplitude, and phase).
- Synthesizing a compensatory sinusoidal signal using the estimated parameters.
- Subtracting this compensatory signal from the original ECG to suppress the interference.

To demonstrate the adaptive behavior under frequency drift (49.5–50.5 Hz range), the estimated power-line frequency was tracked over time. Figure 5 shows the tracking result, where the algorithm successfully follows the stepwise changes in the simulated drifting interference frequency.

After applying the compensatory subtraction, the filtered ECG signal in the time domain is shown in Figure 6. The QRS complexes appear clearer with significantly reduced oscillatory noise compared to the raw signal. The corresponding frequency-domain verification is presented in Figure 7. The spectrum confirms the successful removal of the power-line interference peak, with no significant residual component around 50 Hz.

III. Results and Discussion

This section presents the experimental results and analyzes the performance of the fixed 50 Hz notch filter versus the FFT-based adaptive notch filter, with a focus on their ability to remove 50 Hz power-line interference and the resulting impact on Pan-Tompkins QRS detection accuracy under both fixed and drifting conditions.

The raw noisy ECG signal is heavily corrupted by 50 Hz interference, as seen in the time-domain waveform with strong superimposed oscillations and in the FFT spectrum showing a dominant peak at 50 Hz (see Figs. 1 and 2 in the Methodology section). This noise significantly distorts the QRS complexes and reduces detection reliability.

After applying the fixed 50 Hz notch filter, the time-domain waveform becomes smoother, and the 50 Hz peak in the FFT spectrum is substantially attenuated (Figs. 3 and 4). This confirms the filter's effectiveness when the interference frequency is stable and matches the design center frequency.

The FFT-based adaptive notch filter achieves even better suppression. The frequency tracking plot shows the algorithm continuously estimating and following the simulated stepwise drift (49.4–50.6 Hz), demonstrating its key advantage in handling frequency variations.

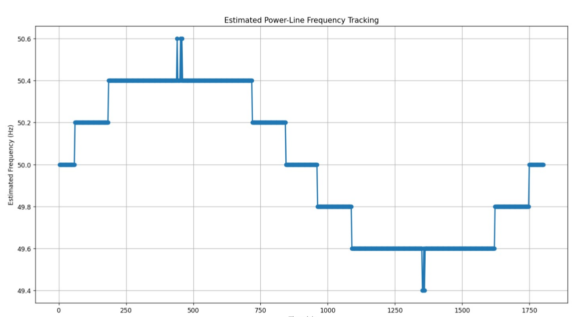

Fig. 5. Estimated power-line frequency tracking over time. The adaptive method successfully detects and adapts to the simulated frequency drift, which fixed filters cannot handle.

The time-domain ECG after adaptive filtering is clean, with QRS complexes preserved and minimal residual oscillations remaining.

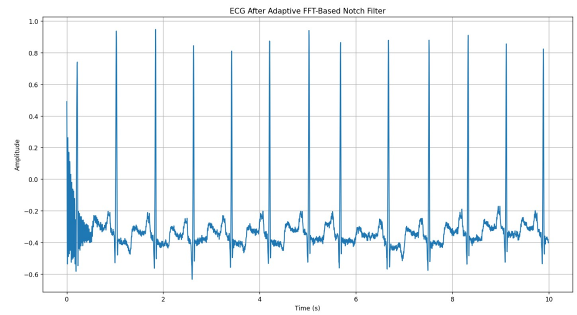

Fig. 6. Time-domain ECG signal after FFT-based adaptive notch filtering. The interference oscillations are effectively removed while QRS morphology is well preserved.

The corresponding FFT spectrum after adaptive filtering confirms near-complete elimination of the 50 Hz peak, even under drift, outperforming the fixed filter.

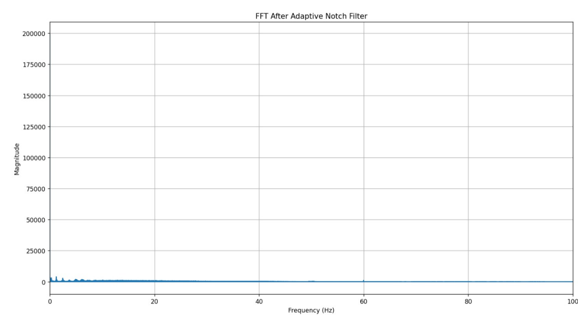

Fig. 7. FFT magnitude spectrum after adaptive notch filtering. The strong 50 Hz interference peak (previously visible in the noisy signal) is almost completely suppressed.

Pan-Tompkins QRS detection on the noisy ECG under drifting interference shows poor performance. The red dots indicate many false positives and missed beats due to residual noise mimicking QRS features.

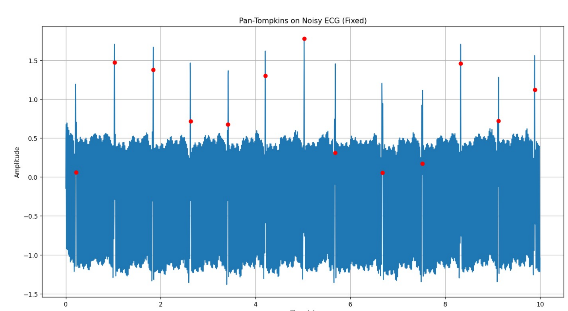

Fig. 8. Pan-Tompkins QRS detection on noisy ECG under drifting interference (red dots: detected R-peaks). Numerous false positives and missed beats occur due to residual noise.

After the fixed notch filter under drifting conditions, detection improves slightly, but residual oscillations still cause several false positives and occasional misses, as shown by the red dots.

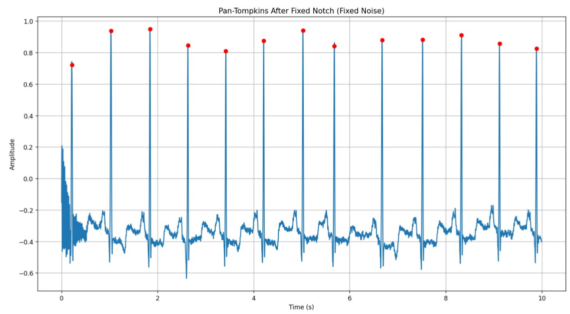

Fig. 9. Pan-Tompkins QRS detection after fixed notch filtering under drifting interference. Some improvement is seen, but false positives persist due to incomplete noise suppression.

In contrast, Pan-Tompkins QRS detection after the adaptive filter is highly accurate. The red dots align well with true R-peaks, with very few false positives and no missed beats, even with drifting interference.

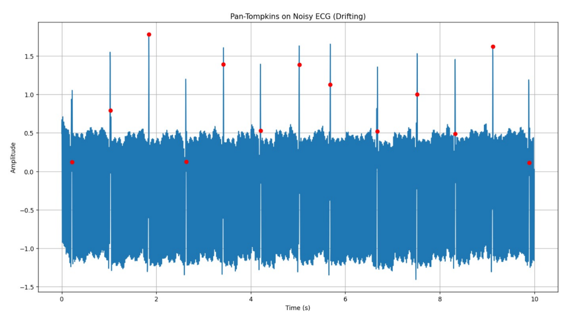

Fig. 10. Pan-Tompkins QRS detection after adaptive notch filtering. Accurate R-peak detection with minimal errors, demonstrating superior robustness to drift.

Under stable (fixed) interference, the fixed notch filter performs reasonably well. Pan-Tompkins detects most R-peaks correctly, with only minor false positives from slight residual noise.

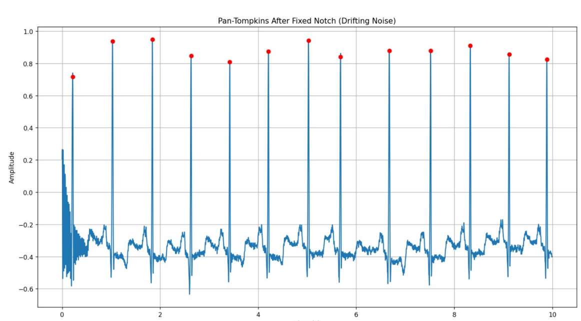

Fig. 11. Pan-Tompkins QRS detection after fixed notch filtering under stable (fixed) interference. Good performance with few errors.

On the noisy signal with fixed interference, detection is unreliable, with many false positives caused by the strong 50 Hz oscillations, as indicated by the red dots.

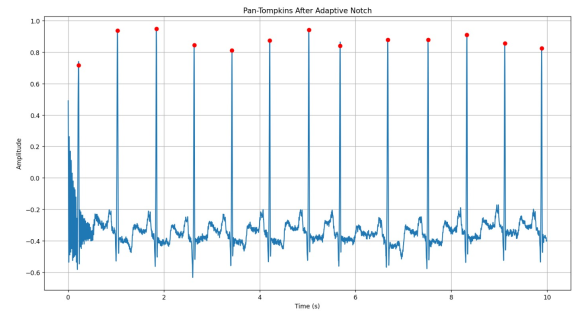

Fig. 12. Pan-Tompkins QRS detection on noisy ECG under fixed interference. High number of false positives due to unfiltered 50 Hz noise.

Overall, the results clearly demonstrate that the FFT-based adaptive notch filter significantly outperforms the conventional fixed filter, especially under drifting interference. It achieves cleaner signals, better preservation of QRS morphology, higher QRS detection accuracy, and greater robustness, making it more suitable for real-world ECG monitoring applications where power-line frequency may vary.

IV. Conclusion

This project investigated the impact of 50 Hz power-line interference on ECG signals and compared two filtering strategies: a conventional fixed notch filter and an FFT-based adaptive notch filter. Both methods were applied to ECG data corrupted with fixed and drifting interference, followed by Pan-Tompkins QRS detection.

The fixed notch filter effectively suppresses interference when the frequency is stable, resulting in improved QRS detection compared to the raw noisy signal. However, under drifting conditions (49.4–50.6 Hz), residual noise persists, leading to increased false positives and missed beats.

The FFT-based adaptive notch filter significantly outperforms the fixed approach. It successfully tracks frequency variations, removes interference more thoroughly, preserves QRS morphology better, and yields higher QRS detection accuracy with fewer errors, even in drifting scenarios.

The results confirm that adaptive filtering is more robust and suitable for real-world ECG monitoring systems, where power-line frequency drift is common. The adaptive method is simple to implement, requires no manual parameter tuning, and provides a practical solution for enhancing signal quality in clinical and ambulatory applications.

Future work could extend the evaluation to real multi-lead ECG datasets, incorporate real-time implementation, or combine the adaptive notch with additional denoising techniques (e.g., baseline wander removal) for even better performance.

References

[1] J. Pan and W. J. Tompkins, “A Real-Time QRS Detection Algorithm,” *IEEE Trans. Biomed. Eng.*, vol. BME-32, no. 3, pp. 230–236, Mar. 1985.

TABLE I
COMPARISON OF FILTERING METHODS AND QRS DETECTION PERFORMANCE

Metric	**Noisy (Drift)**	**Fixed Notch (Drift)**	**Adaptive Notch**
Residual 50 Hz Power	High	Medium	Very Low
SNR Improvement	-	Moderate	High
QRS Accuracy (%)	Low	Medium	High
False Positives	Many	Several	Very Few
Missed Beats	Several	Few	None
Robustness to Drift	Poor	Limited	Excellent

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